QINYU DONG

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Research: Embodied AI; Physical Intelligence; RL; World Models; Agentic RL

EDUCATION

2024.09-Present Tsinghua Univ. (985/211/Double First-Class) Dept. of Automation Master's Student, Control Science & Engineering

2018.09-2024.07 (Military Service: 2019-2021) Nanjing Univ. of Aeronautics & Astronautics (211/Double First-Class) College of Astronautics B.Eng., Aircraft Control & Information Engineering

- **GPA: 4.1/5.0 Major Rank: 1/43**
- **Selected Coursework:** Automatic Control (99), Modern Control (99), System Identification (100), Information Fusion (99), Optimal Control (92), Convex Optimization (90).
- **Honors:** University Achievement Award; President's Special Commendation; National Scholarship; Provincial Merit Student; Tsinghua Comprehensive Excellence Scholarship (Second Class); Outstanding Student Award (1st); Academic Scholarship (1st); University Merit Student.
- **English & Certification:** CET-4: 611; CET-6: 503; MIIT Certificate in Mathematical Modeling.

PAPERS

2026.05*-Present** ***Teacher-Free Semantic Distillation for V-JEPA 2 Latent World Models (Manuscript in Preparation; Targeting ICLR 2027) **Second Author**

Context: Existing latent world models lack high-level semantic understanding, limiting state prediction and decision-making in complex robotic tasks. Proposed a VLM-supervised world-model enhancement framework, ***PrivilegedJEPA***.

Contributions:

1. Designed a teacher-inference-free semantic distillation framework that injects multimodal priors from a frozen vision-language model into a latent world model.
2. Introduced cosine alignment loss and reference predictor retention loss to enhance semantics while preserving predictive capability.
3. Trained and evaluated the Stable World Model on Push-T, Two-Room, Reacher, and OGBench-Cube robotic-control tasks.

Outcome: Achieved a 90% closed-loop success rate; improved Two-Room, Reacher, and OGBench-Cube by 8, 4, and 6 percentage points; prepared a manuscript targeting ***ICLR 2027***.

2026.03*-2026.05** ***G-EATS: A Goal-Conditioned RL Framework with Energy-Aware Target Streams for Agile Fixed-Wing Maneuvering (CoRL 2026) **First Author**

Context: Traditional control for agile fixed-wing maneuvers depends on hand-crafted dynamics and struggles with complex trajectory planning. Proposed the goal-conditioned RL flight-control framework ***G-EATS***.

Contributions:

1. Designed a goal-conditioned RL policy for direct actuator commands in quaternion state space.
2. Developed energy-aware PPO and selected task-specific target-stream parameters via closed-loop rollouts to improve agile tracking and energy safety.
3. Built a high-fidelity fixed-wing dynamics simulator and conducted multi-task, multi-scenario RL training and evaluation.

Outcome: Demonstrated stable closed-loop behavior across multiple simulated fixed-wing maneuvers. Open-sourced code: https://anonymous.4open.science/r/Energy-Aware-Target-Streams-for-Learned-Fixed-Wing-Maneuvering-B550/**; published in *CoRL 2026***.

2025.03-2026.03 ***Planax: An Efficient High-Fidelity Benchmarking Platform for Large-Scale UAV Swarm Cooperative Combat*** **(*IEEE RA-L*)** **Co-First Author**

Context: Led development of Planax, a JAX-based high-fidelity parallel simulator addressing the fidelity-efficiency trade-off in RL training for large fixed-wing UAV swarms.

Contributions:

1. Developed a high-fidelity JAX/XLA parallel UAV simulator with end-to-end dynamics.
2. Implemented and benchmarked PPO, MAPPO, and hierarchical RL baselines; completed self-play training for 50-vs-50 UAV swarms.

Outcome: Achieved >10x simulation throughput over NeuralPlane under the reported benchmark with lower GPU memory use. Fully open-sourced: https://anonymous.4open.science/r/Aeroplanax-2087**; published in *IEEE RA-L***.

2024.09-2025.03 ***Research on the Control Algorithms for an Attitude Control System with 4-Axis Reaction Wheels of Pyramid Configuration*** **(Accepted by *CCC 2025*; EI-indexed)** **First Author**

Context: Proposed a fault-tolerant, redundant four-wheel pyramid configuration to address low space utilization and high power consumption in conventional orthogonal three-wheel CubeSat systems.

Contributions:

1. Derived three-axis dynamics and the pseudoinverse torque-allocation matrix; built a coupled multibody simulation in SolidWorks and Simscape.
2. Designed and compared PID/LQR/SMC controllers; completed full-stack hardware/software development and closed-loop integration using STM32G431 and 3D printing.

Outcome: Demonstrated 90-degree attitude maneuvers and robustness under a 150 mN • m disturbance. Prototype video received **28K+ views on Bilibili** (https://www.bilibili.com/video/BV1XBTNeeEMK/). The **first-authored work** was **accepted by *CCC 2025*** **(EI)**.

2023.10-2024.05 **Control Algorithms for a Pyramid-Configuration Four-Reaction-Wheel Attitude Control System for Micro/Nanosatellites** **Undergraduate Thesis**

Contributions:

Modeled four-wheel pyramid dynamics; designed PID, LQR, and sliding-mode controllers; conducted multibody dynamics simulations in Simscape.

Outcome: Grand Prize, National Undergraduate Thesis Exchange for Aeronautics and Astronautics; Second Prize, University Outstanding Thesis.

INDUSTRY

2026.07-Present ByteDance

Project: Doubao Agent Skill Mining and Evaluation

Contributions:

1. Built an Agent Trace-based framework for Skill demand mining; designed Task Unit extraction, demand clustering, and capability-matching algorithms.
2. Developed Skill Capability representations and an evaluation system; analyzed coverage, overlap, and failure attribution.
3. Provided recommendations for optimizing, creating, or integrating Agent Skills to inform tool selection and capability planning.

RESEARCH

2026.02*-Present** ***AutoPlanax: Agentic Framework for Automated RL Research **Project Lead**

Context: RL reward design relies heavily on manual trial and error. Drawing on Eureka/auto-research paradigms, built AutoPlanax, an LLM-agent-driven automated RL experimentation framework on Planax.

Contributions:

1. Designed an LLM-agent RL experimentation framework with staged permission controls;

2. Developed reward reflection and in-context evolutionary search;

3. Implemented early exit, automatic Git rollback, and other automation modules.

Outcome: Completed 68 automated experiments; reduced the best-performing policy's tracking error from >90 degrees to 20 degrees (geodesic angle in quaternion space); established an auto-research loop of "LLM hypothesis -> RL training -> automated evaluation -> history-aware feedback."

AWARDS

2023.03-2023.08 **Air-Bearing CubeSat Attitude Control with Three Momentum Wheels** **Team Member**

Contributions:

1. Implemented MPU6050 attitude estimation (DMP), Bluetooth remote attitude control, and signal tracking (STM32CubeMX, Keil 5).
2. Designed and tuned a PID controller; designed the CubeSat motor-driver PCB (Altium Designer); developed host software in MATLAB.

Outcome: 6th National Undergraduate Embedded Chip and System Design Competition: **National Second Prize**; **First Prize, Eastern Regional Final**.

2022.04-2023.04 **Intelligent Logistics Vehicle Platform Based on an Open-Source Hardware Board** **Team Lead**

Contributions:

1. Developed the STM32 board and vision system (STM32CubeMX, Keil 5, OpenMV IDE).
2. Improved Otsu and particle-swarm methods; designed control laws; performed kinematic modeling and simulation in MATLAB.
3. Designed PCBs for all modules (EasyEDA, Altium Designer) and built the physical prototype.

Outcome: Rated Outstanding in a national undergraduate innovation project; **one software copyright and one invention patent granted**; **first-authored paper** published in **Industrial Control Computer**.

2022.02 ***Destruction and Regeneration: Carbon Sequestration in a Simulated Forest*** **Team Lead**

2022.09 **Chemical Composition and Classification of Ancient Glass Products** **Team Lead**

Contributions:

1. Data mining and modeling: mathematical models, cellular automata, analytic hierarchy process, and machine learning (decision trees, random forests).
2. Modeling and programming: deep-learning algorithm analysis.

Outcome: Interdisciplinary Contest in Modeling (ICM): **International Second Prize**; China Undergraduate Mathematical Contest in Modeling: **Provincial Third Prize**.

PROJECTS

2025.05-2025.10 **RL & High-Fidelity Flight-Simulation Data Toolchain (Industry-Sponsored)** **Project Lead**

Contributions:

1. Extended the SMAC/PyMARL training pipeline to extract complete Markov transition tuples.
2. Developed AeroPlanax log parsing to recover normalized 6-DOF observations into physical quantities (angle of attack, attitude, etc.) and generate reports.
3. Developed a C#/WPF graphical data plug-in with the Tacview SDK for bidirectional multi-format import/export and validation.

Outcome: Built an end-to-end "raw data extraction - feature parsing - 3D visualization/replay (Tacview)" pipeline, standardizing and automating large-scale simulation data management, accelerating experimental iteration, and supporting paper development.

SERVICE

2019.09-2021.09 **Military Service** **Infantry**

- Completed training for the **"Sniper Frontier 2021" international military competition** and competed in brigade-, battalion-, and company-level combat-skill events; received **12 service and training honors**, including Outstanding Recruit, Training Pacesetter, and Outstanding Marksman, plus **two brigade commendations**.

- **Outstanding Volunteer** (160 hours); 5th place in pull-ups and **Team First Prize** at the 60th university athletics meet.

SKILLS

- **AI/Agent Tools:** Proficient with Claude Code and the Anthropic API for agentic workflow development; familiar with verl.
- **Tools:** Git, VS Code, PyCharm, Simulink, Keil, Isaac Gym, STM32CubeMX, SolidWorks.
- **Programming:** Python, MATLAB, C/C++; Frameworks: JAX, PyTorch, TensorFlow.